

Distribution	Formula	Beta Parameter 1	Beta Parameter 2	Notes
		Normal distribution		standard deviation of the corresponding Normal distribution, sigma.
Uniform	$1/(B - A)$ for $A < x < B$	A	B	The Uniform distribution has two parameters, A and B, which define the interval on which the distribution is defined.
Exponential	$\frac{1}{\mu} e^{-x/\mu}$	Mu	N/A	The Exponential distribution has one parameter, mu.
Geometric	$p(1 - p)^x$ for $0 < p < 1$	p	N/A	The Geometric distribution has one parameter, p.
Weibull	$\left(\frac{\beta}{\alpha}\right)\left(\frac{x}{\alpha}\right)^{\beta-1} e^{-(x/\alpha)^\beta}$	alpha	beta	The Weibull distribution has two parameters, alpha and beta.
Gamma	$\frac{1}{b^a \Gamma(a)} x^{a-1} e^{-x/b}$	a	b	The Gamma distribution has two parameters, a and b.
Logistic	$\frac{e^{(x-m)/b}}{b \left[1 + e^{(x-m)/b}\right]^2}$	m	b	The Logistic distribution has two parameters, m and b.
Beta	$\frac{1}{\beta(a, b)} x^{a-1} (1 - x)^{b-1}$	a	b	The Beta distribution has two parameters, a and b.
Pareto	ab^a / x^{a+1}	a	b	The Pareto distribution has two parameters, a and b.
Cauchy	$\frac{1}{\pi} \cdot \frac{b}{(x - m)^2 + b^2}$	b	m	The Cauchy distribution has two parameters, b and m.
Custom	N/A	Standard deviation	N/A	The custom distribution is used for specified expert distributions for pollutant effect (e.g., truncated parametric distributions or non-parametric distributions).

For the **Function** definition, commonly used mathematical operators (+, -, *, /) may be used. Other available “**Operators**” are listed to the right of the commonly used functional forms (e.g., ABS(x), EXP(x), LOG(x)). These operators are supported by the math and statistics library used within BenMAP-CE.

Under the heading “**Available Variables**”, are temporary runtime variables which can be used in the health impact function definition. The values for these variables (except for the A, B, C constants) are expected to change as the code loops over grid cells and stratified population groups. Table 4-11 presents the available runtime variables that can be used in health impact function definition.

Table 4-11. Health Impact Function Available Runtime Variables

Variable	Type	Notes
Beta	Numeric	Beta coefficient
DELTAQ	Numeric	Change in pollutant concentration (baseline – control)
POP	Numeric	Population
Incidence	Numeric	Number of people with new (or newly diagnosed) adverse health effects within a given period of time.
Prevalence	Numeric	Number of people who already have a given adverse health condition (i.e., chronic illness).
Q0	Numeric	Control pollutant concentration
Q1	Numeric	Baseline pollutant concentration
A	Numeric	User-defined constant
B	Numeric	User-defined constant
C	Numeric	User-defined constant

The “**Population Variables (optional)**” box provides a list of setup variables which have been defined under the **Variable Datasets** (see **Section 4.1.7**).

4.1.7 Variable Data

Health Impact Functions and **Valuation Functions** may sometimes refer to socioeconomic variables for which BenMAP-CE does not automatically calculate values.

Fundamental Concept – Variable Data

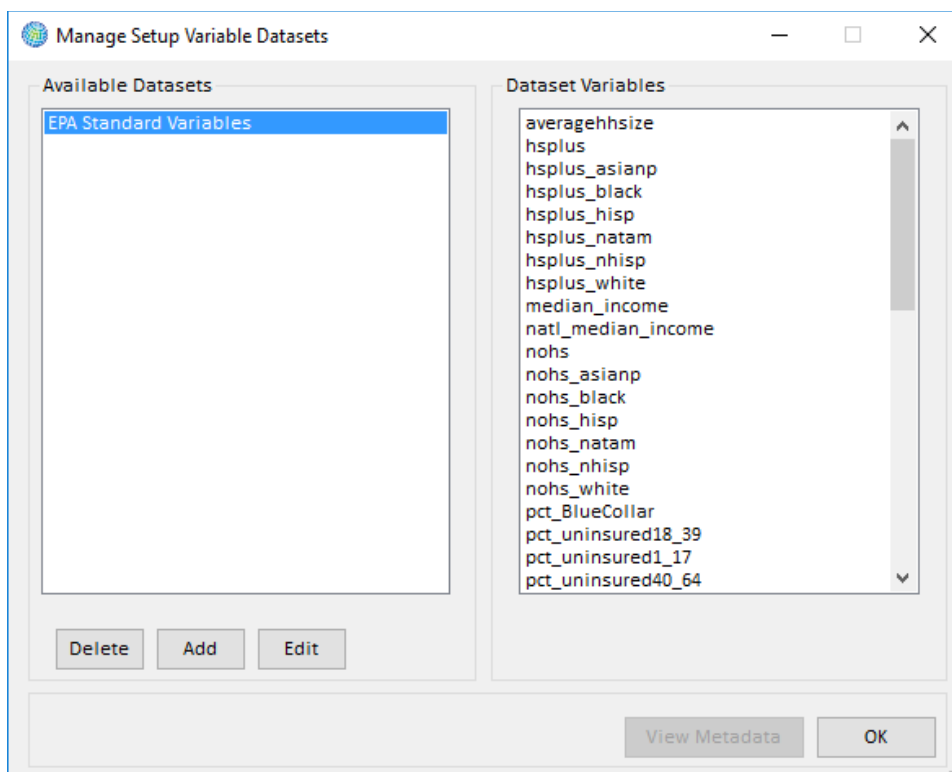
A **Variable Data** dataset contains sociodemographic and economic data not contained elsewhere that may be needed for certain health impact or valuation functions, or for certain types of analysis. Examples of socioeconomic variables that might be included in **Variable Data** include educational attainment and percentage of population living below the poverty line.

Variables included in the dataset must be associated with a particular **Grid Definition**. Data at different spatial scales (e.g., county versus state) need to be input separately using different Variable Data files, each associated with the relevant grid definition.

For example, some valuation functions reference the wage (including fringe benefits and overhead) within each area of analysis. Other functions apply to populations living below the poverty line in a given country. To facilitate this type of analysis, BenMAP-CE allows you to load datasets of socioeconomic **Variables**, which apply either globally or to a specific geographic area (i.e., they are associated with a **Grid Definition**).

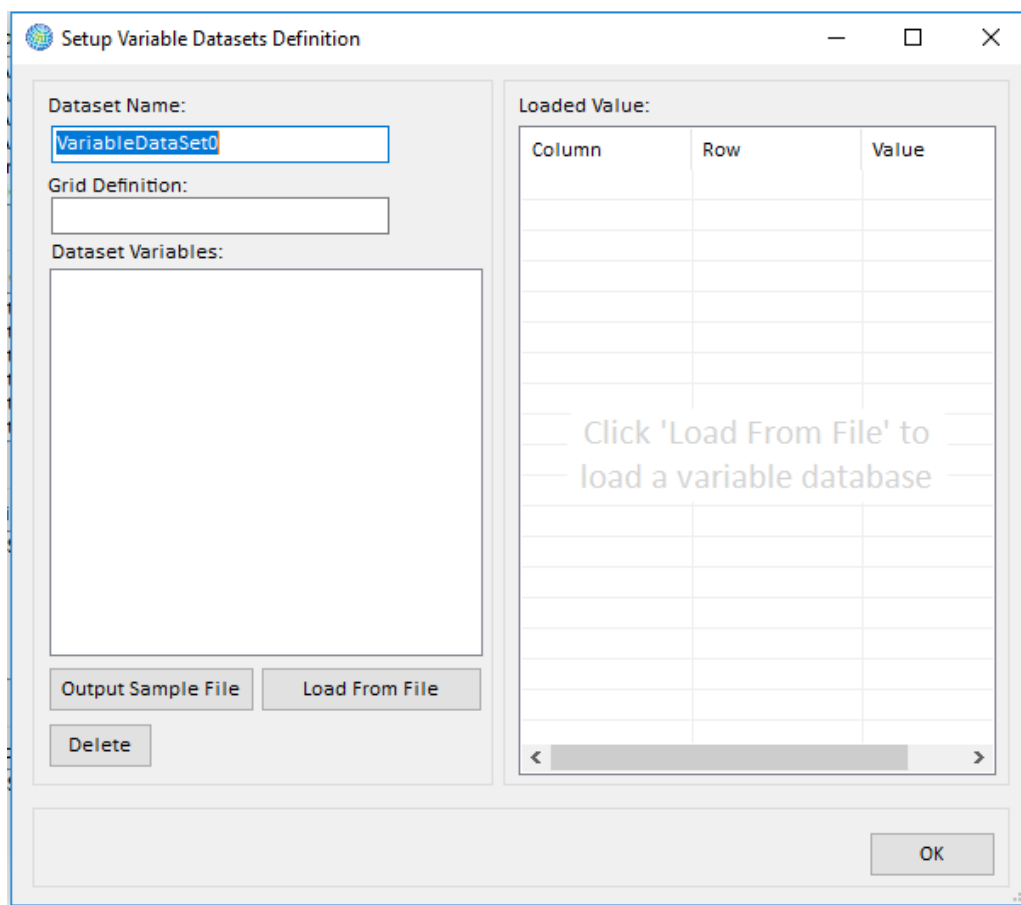
4.1.7.1 Add Variable Data

To add **Dataset Variables** to BenMAP-CE (such as income and other miscellaneous variables that might be needed in the analysis), click on the **Manage** button below the **Variables Datasets** box in the **Modify Datasets** window. The **Manage Setup Variable Datasets** window will appear.



In this window you may **Add**, **Edit**, and **Delete** datasets. The section on the left under **Available Datasets** lists the variables datasets that are currently in the setup database. The section on the right under the **Dataset Variables** lets you view the variables in a selected dataset.

To add a **Variable** dataset click the **Add** button. This will take you to the **Setup Variable Dataset Definition** window. In this window you may add externally created variables through the **Load From File** button for any of your predefined **Grid Definitions**.



To start, type the name that you want to use for the **Variable Dataset** in the **Dataset Name** box. (This is a name that is internal to BenMAP-CE and used just for identification.)

To add an externally created **Variable Dataset**, click the **Load From File** button. This will bring up the **Load Variable Database** window. Here you need to choose the grid definition from the **Grid Definition** drop-down list that matches the level of aggregation in your variable data file. Remember that the **Variable Dataset** you import must use the same column/row index as the **Grid Definition**. Next, you may use the **Browse** button to find and select the desired **Database** (i.e., input file) and click **Open**. You can click the **Validate** button to ensure the file is properly formatted before importing. BenMAP-CE's validation tool will review the file format (column names, required columns, and data types) and provide a report with any errors or warnings. You can also add metadata to the imported data (reference and description of the data file) by clicking the **View Metadata** button.